BJT DC bias II

Answers and worked solutions · Day 08 · Sat 31 Oct · pack E5

5. Practice

From your course files

WS5 Q12 (course question).

Part 1. The base current is fixed by the source: I_B = 0.01 mA.

Guess active. $I_C = \beta I_B = 150(0.01) = 1.5$ mA and $I_E = I_C + I_B = 1.51$ mA. Collector loop:

$$V_{CE} = 12 - 9I_C - 1 \cdot I_E = 12 - 13.5 - 1.51 = -3.01 \text{ V}$$

Negative, so not active.

Guess saturation. V_{CE} = 0.2 V, and I_B is still 0.01 mA, so

$$I_E - I_C = 0.01 \quad (A), \qquad 9I_C + 1 \cdot I_E = 11.8 \quad (B)$$

Solving: I_C = **1.179 mA**, I_E = **1.189 mA**.

Check: $I_B, I_C > 0$ and $I_C = 1.179 < \beta I_B = 1.5$ mA (the solution writes this as $I_B = 0.01 > I_C/\beta = 0.00786$ mA). **Saturated**, forced β = 117.9.

Part 2. At the edge of saturation $I_C = \beta I_B$ and $V_{CE} = 0.2$ V, which gives $I_B[\beta(9 + R_E) + R_E] \geq 11.8$. The worst case is $\beta_{\min}$ = 100: $101R_E \geq 1180 - 900$, so $R_E \geq$ **2.77 kΩ**. The full argument, and why the reverse question ("always active") has no answer, is in Example 4 above.

WS5 Q13 (course question).

The emitter is grounded, so in saturation $V_C = V_{CE,sat} = 0.2$ V and

$$I_{C,sat} = \frac{5 - 0.2}{1 \text{ k}\Omega} = 4.8 \text{ mA}$$

At the edge of saturation $I_B = I_{C,sat}/\beta$.

Course answer (uses $\beta_{\max}$ = 200): $I_{B,\min} = 4.8/200$ = **24 µA**, and the base loop gives $V_{BB} = 0.7 + 100(0.024)$ = **3.1 V**.

Read this carefully. 24 µA is the current at which a β = 200 device *starts* to saturate. A β = 100 device needs $4.8/100$ = **48 µA**, which is $V_{BB} = 0.7 + 100(0.048)$ = **5.5 V**. To

guarantee saturation for every device in the range, give 48 µA and 5.5 V. In a test, give the course's number and add this line, saying which one you mean.

WS5 Q17 (course question).

V_{CE} = 1 V > 0.2 V, so the BJT is active.

Base loop (10 V, 80 kΩ, V_{BE}, 5 kΩ): the 80 kΩ goes straight from the 10 V rail to the base, so the base loop does not involve R_C:

$$I_B = \frac{10 - 0.7}{80 + 5(101)} = 15.9\ \mu\text{A}, \qquad I_E = 101 I_B = 1.606\ \text{mA}, \qquad I_C = 100 I_B = 1.590\ \text{mA}$$

Collector side. Thevenin the collector network (R_C from 10 V, 100 kΩ to ground): $V_{Th} = 10 \cdot \dfrac{100}{100 + R_C}$ and $R_{Th} = \dfrac{100 R_C}{100 + R_C}$. With $V_C = V_{CE} + 5 I_E = 1 + 8.03 = 9.03$ V:

$$V_{Th} - R_{Th} I_C = 9.03 \ \Rightarrow\ 1000 - 158.97 R_C = 9.03(100 + R_C) \ \Rightarrow\ R_C = \frac{97.2}{168.0} = \mathbf{0.58\ k\Omega}$$

The course solution prints **0.59 kΩ**, because it rounds V_C = 9.03 V to 9 V in the last line ($100 = 9 R_C + 158.9 R_C$). Either is fine; 0.58 kΩ is the unrounded value.

WS5 Q18 (course question).

V_{CE} = 6 V, so the BJT is active. $I_E = I_C/\alpha = 2.030$ mA and $I_B = I_E - I_C$ = 0.030 mA (β = α/(1−α) = 65.7).

$$V_B = V_{BE} + 2 I_E = 0.3 + 2(2.030) = 4.36\ \text{V}$$

Course method (divider, base current ignored): $4.36 = 16 \cdot \dfrac{20}{20 + R_1}$, so R_1 **= 53.4 kΩ**.

Collector loop: $16 = 2 R_C + 6 + 2 I_E$, so $R_C = (16 - 6 - 4.06)/2$ = **2.97 kΩ**.

Why the course's R_1 is only approximate. βR_E = 65.7 × 2 kΩ = 131 kΩ, which is less than $10 R_2$ = 200 kΩ, so the "ignore I_B" approximation fails its own test. Exactly, R_1 carries $V_B/20\text{k} + I_B$ = 0.218 + 0.030 = 0.248 mA, so $R_1 = (16 - 4.36)/0.248$ = **46.8 kΩ**. If you are asked this in a test, give the exact value and say why.

WS5 Q19 (course question).

V_{CE} = 2 V, so active: $I_B = I_C/\beta$ = 0.01 mA and I_E = 1.01 mA.

Collector feedback: R_C carries $I_C + I_B = I_E$. The emitter is grounded, so $V_C = V_{CE}$ = 2 V:

$$R_C = \frac{15 - 2}{1.01} = 12.87\ \text{k}\Omega$$

(The question asks only for R_B, but you need R_C first. The course solution finds both.)

Base path: $15 - R_C I_E - R_B I_B - V_{BE} = 0$, so

$$R_B = \frac{15 - 0.3 - 12.87(1.01)}{0.01} = \mathbf{170\ k\Omega}$$

Quicker: $R_B = (V_C - V_{BE})/I_B = (2 - 0.3)/0.01$ = 170 kΩ.

WS5 Q9 (course question, *).

The collector is tied to +20 V, so V_C = 20 V. Guess active.

Base loop: $-20 + 10I_B + 0.7 + 1(\beta + 1)I_B = 0$, so

$$I_B = \frac{19.3}{10 + 101} = \mathbf{0.174\ mA}$$

$V_B = 20 - 10(0.174) = 18.26$ V and $V_E = V_B - 0.7 = 17.56$ V, so $V_{CE} = 20 - 17.56$ = **2.44 V**.

Check: $I_B > 0$ and V_{CE} > 0.2 V, so **active**. (I_C = 17.4 mA, I_E = 17.56 mA.)

WS5 Q11 (course question, *).

Guess active. Base-emitter loop: $V_B - 0.7 - 1 \cdot I_E = 0$, so $I_E = (0 - 0.7)/1 = -0.7$ mA. With very high β, $I_C \approx I_E = -0.7$ mA, so I_B would also be negative. (The collector loop gives $V_{CE} = 3 - 1(-0.7) - 1(-0.7) = 4.4$ V, but that does not rescue it.) $I_B < 0$, so not active.

Guess cut-off. $I_B = I_C = I_E = 0$, so V_E = 0 and $V_{BE} = 0 - 0 = 0$ V < 0.7 V. **Cut-off.** (V_C = 3 V.)

WS5 Q14 (course question, *).

Thevenin the base: $V_{Th} = 20 \cdot \frac{10}{100}$ = 2 V and R_{Th} = 90 ∥ 10 = 9 kΩ. Guess active:

$$I_B = \frac{2 - 0.7}{9 + 51(1)} = \mathbf{21.7\ \mu A}, \qquad I_C = 50 I_B = \mathbf{1.08\ mA}, \qquad I_E = 1.105\ \text{mA}$$

$$V_{CE} = 20 - 5I_C - 1 \cdot I_E = 20 - 5.42 - 1.11 = \mathbf{13.5\ V}$$

$I_B > 0$ and V_{CE} > 0.2 V, so **active**.

WS5 Q15 (course question, *).

Two Thevenins.

- Base: $V_{Th,b} = V_I \cdot \dfrac{40}{400} = 0.1V_I$ = 1.5 V, and $R_{Th,b}$ = 360 ∥ 40 = 36 kΩ.
- Collector (superposition of +5 V and −10 V): $V_{Th,c} = 5 \cdot \dfrac{27}{30} - 10 \cdot \dfrac{3}{30}$ = 3.5 V, and $R_{Th,c}$ = 3 ∥ 27 = 2.7 kΩ.

Guess active (emitter grounded):

$$I_B = \frac{1.5 - 0.7}{36} = 22.2\ \mu\text{A}, \qquad I_C = 40I_B = 0.889\ \text{mA}$$

$$V_o = V_C = 3.5 - 2.7(0.889) = \mathbf{1.10\ V}$$

$V_{CE} = V_o$ = 1.10 V > 0.2 V and $I_B > 0$, so **active**.

The course solution gives 1.124 V because it rounds I_B to 22 µA (I_C = 0.88 mA). Either is fine.

PT Q8 (practice test, 6 marks).

READING THE FIGURE. Q_1 is an npn whose base and collector are joined at the 2.5 V terminal. Its emitter connects directly to the emitter of Q_2, a pnp. Q_2's base goes through 10 kΩ to the 1 V terminal, and its collector is the output V_{C2}, with 1 kΩ to ground.

Step 1, Q_1. With base tied to collector, $V_{CB1} = 0$, so Q_1 is diode-connected and always active. It drops one V_{BE}:

$$V_{E1} = 2.5 - 0.7 = 1.8\ \text{V} = V_{E2}$$

Step 2, Q_2 base (pnp). Emitter above base:

$$V_{B2} = V_{E2} - V_{EB2} = 1.8 - 0.7 = 1.1\ \text{V}$$

Step 3, base current. For a pnp the base current flows out of the base, from 1.1 V through 10 kΩ to the 1 V terminal:

$$I_{B2} = \frac{1.1 - 1.0}{10\ \text{k}\Omega} = 10\ \mu\text{A}$$

Step 4, collector. $I_{C2} = \beta_2 I_{B2} = 50 \times 10\ \mu\text{A}$ = 0.5 mA. It flows out of the collector, down through 1 kΩ to ground:

$$V_{C2} = 0.5\ \text{mA} \times 1\ \text{k}\Omega = \mathbf{0.5\ V}$$

Step 5, checks. $V_{EC2} = 1.8 - 0.5 = 1.3$ V > 0.2 V and I_{B2} > 0, so Q_2 is active. $I_{E2} = 51 \times 10\ \mu\text{A}$ = 0.51 mA. That current flows out of Q_1's emitter, so I_{B1} = 0.51/101 =

5.05 µA and I_{C1} = 0.505 mA, both drawn from the 2.5 V terminal. These are consistent but play no part in V_{C2}. **β1 does not affect the answer**, because Q_1 only fixes the 1.8 V level. Say this in one line. It agrees with the Test 1 Reference.

PT Q7 (practice test, 10 marks).

(a) With the base disconnected, the divider runs from +9 V to −12 V:

$$V_{Th} = 9 \cdot \frac{30}{45} + (-12) \cdot \frac{15}{45} = 6 - 4 = \mathbf{2\ V}, \qquad R_{Th} = 15 \parallel 30 = \mathbf{10\ k\Omega}$$

(b) The drawing: V_{Th} = 2 V in series with R_{Th} = 10 kΩ into the base; a 0.7 V source from base to the grounded emitter; a current source βI_B from collector to emitter; R_C = 1.5 kΩ from V_{CC} = 12 V to the collector.

(c) Emitter grounded, so

$$I_B = \frac{2 - 0.7}{10} = 0.130\ \text{mA}, \qquad I_C = 50 I_B = \mathbf{6.5\ mA}, \qquad V_C = 12 - 1.5(6.5) = \mathbf{2.25\ V}$$

$V_{CE} = 2.25$ V > 0.2 V, so the active assumption holds. This matches the Test 1 Reference.

The approximate method, compared with the exact one.

Step 1, validity test. The approximate method needs $\beta R_E \geq 10 R_2$. Here $R_E = 0$, so $\beta R_E = 0 < 10 \times 30$ kΩ = 300 kΩ. **The test fails completely.**

Step 2, what happens if you try anyway. The approximate method takes $V_B = V_{Th}$ = 2 V, so $V_E = 2 - 0.7 = 1.3$ V. But the emitter is grounded, so V_E must be 0. With no R_E, the "1.3 V across R_E" step would give $I_E = 1.3/0 \to \infty$. The method has nothing to work with. It relies on R_E to set the current, and this circuit has none.

Step 3, why, with numbers from the exact solution. The base is clamped at $V_B = V_{BE}$ = 0.7 V, not 2 V:

- current in R_1: $(9 - 0.7)/15$ = 0.553 mA
- current in R_2: $(0.7 + 12)/30$ = 0.423 mA
- base current: 0.553 − 0.423 = **0.130 mA**, which agrees with the exact method.

The base current is 24 % of the divider current, so it is far from negligible. That is exactly what the $\beta R_E \geq 10 R_2$ test is designed to detect.

Comparison where the approximate method does give a number (WS5 Q7, the divider-bias circuit from E4): βR_E = 54.4 kΩ < $10 R_2$ = 91 kΩ, so the test fails, but only mildly.

	Exact	Approximate	Error
I_C	1.71 mA	1.98 mA ($I_E = (2.048 - 0.7)/0.68$)	+16 %
V_{CE}	8.15 V	6.92 V	−15 %

For the lecture example (22 V, 39 kΩ / 3.9 kΩ, R_E = 1.5 kΩ, β = 100), where the test passes, the approximate method gives I_C = 0.867 mA and V_{CE} = 12.03 V. The exact method gives 0.838 mA and 12.36 V (slide 33 prints 0.84 mA and 12.34 V). The error is about 3 %.

Conclusion to write: the approximate method is only as good as $\beta R_E/R_2$ is large. It cannot be used at all without an emitter resistor, so practice test Q7 must be done with Thevenin.

New questions

New Q1 [5 marks].

(a) Active.

$$I_B = \frac{5 - 0.7}{47 + 121(0.5)} = \frac{4.3}{107.5} = 40.0\ \mu\text{A}, \qquad I_C = 120 I_B = 4.80\ \text{mA}, \qquad I_E = 4.84\ \text{mA}$$

$$V_{CE} = 9 - 4.80(2.2) - 4.84(0.5) = -3.98\ \text{V}$$

This is negative, so the transistor is not active.

(b) Saturation. Set V_{CE} = 0.2 V and $I_E = I_B + I_C$. Because R_E is shared, the two loops are coupled:

- Base loop: $5 = 47 I_B + 0.7 + 0.5(I_B + I_C)$, so $47.5 I_B + 0.5 I_C = 4.3$.
- Collector loop: $9 = 2.2 I_C + 0.2 + 0.5(I_B + I_C)$, so $0.5 I_B + 2.7 I_C = 8.8$.

Solving: I_B **= 56.3 µA and** I_C **= 3.25 mA. Check:** both are positive, and I_C = 3.25 mA < βI_B = 6.76 mA. **Saturated.** The forced β is 3.25/0.0563 = **57.7**.

I_B changed from attempt 1 to attempt 2 because R_E links the base loop to the collector current. The shortcut "I_B unchanged" only works when the emitter is grounded.

New Q2 [6 marks] (pnp).

(a) I_E flows from +10 V through R_E *into* the emitter. I_B flows *out* of the base, down through R_B to ground. I_C flows *out* of the collector, down through R_C to ground. KVL from +10 V through R_E, the emitter-base junction and R_B to ground:

$$10 - 2I_E - 0.7 - 220 I_B = 0, \quad I_E = 81 I_B \Rightarrow I_B = \frac{9.3}{220 + 81(2)} = \frac{9.3}{382} = \mathbf{24.3\ \mu A}$$

$$I_C = 80 I_B = \mathbf{1.948\ mA}, \qquad I_E = 1.972\ \text{mA}$$

(b) $V_E = 10 - 2(1.972) = 6.06$ V. $V_B = V_E - 0.7 = 5.36$ V, which equals 220×0.0243, a useful check. $V_C = 2.2 \times 1.948 = 4.28$ V. $V_{EC} = 6.06 - 4.28 = \mathbf{1.77\ V} > 0.2$ V and $I_B > 0$, so **active**.

(c) At the edge, $V_C = V_E - 0.2 = 5.86$ V with the same I_C, because the base loop does not involve R_C. So $R_{C,\max} = 5.86/1.948 = \mathbf{3.01\ k\Omega}$.

New Q3 [7 marks] (sensitivity).

(a) $I_B = 4.3/100$ = 43 μA for both values of β, because the emitter is grounded.

- β = 100: I_C = 4.30 mA and $V_{CE} = 12 - 4.3(2.2)$ = 2.54 V. **Active.**
- β = 200: in the active model I_C = 8.6 mA, which gives V_{CE} = −6.92 V, so the transistor is not active. In saturation, $I_C = 11.8/2.2$ = **5.36 mA** (< βI_B = 8.6 mA). **Saturated.**

(b) The worst case is $\beta_{\max}$ = 200, because it gives the most collector current. Active requires

$$V_{CE} = 12 - I_B\left[\beta R_C + (\beta+1)R_E\right] \geq 0.2, \qquad I_B = \frac{4.3}{100 + (\beta+1)R_E}$$

Multiplying out: $11.8[100 + 201R_E] \geq 4.3[440 + 201R_E]$, so $201R_E(7.5) \geq 1892 - 1180 = 712$. That gives $R_E \geq$ **0.472 kΩ**.

(c) The E12 value 470 Ω is just *below* the limit. At β = 200 it gives V_{CE} = 0.18 V, which is saturated. So choose **560 Ω**:

β	I_B	I_C	V_{CE}
100	27.5 μA	2.75 mA	4.40 V
200	20.2 μA	4.05 mA	0.82 V

I_C now changes by 47 % over the range instead of 100 %. The circuit is still not well stabilised, because $(\beta+1)R_E$ is only about the same size as R_B. That is why practical designs use a divider (New Q5).

New Q4 [8 marks] (two transistors, npn into pnp).

(a) $\beta_1 R_{E1}$ = 150 kΩ ≥ $10R_2$ = 120 kΩ, so the approximate method is valid. Exact: R_{Th} = 56 ∥ 12 = 9.88 kΩ and E_{Th} = 12(12/68) = 2.118 V.

$$I_{B1} = \frac{2.118 - 0.7}{9.88 + 151(1)} = 8.81\ \mu\text{A}, \qquad I_{C1} = \mathbf{1.322\ mA}, \qquad I_{E1} = 1.331\ \text{mA}$$

(b) $V_{C1} = 12 - 1.322(4.7)$ = 5.79 V. For the pnp, $V_{E2} = V_{B2} + 0.7$ = 6.49 V, and $I_{E2} = (12 - 6.49)/2.2$ = 2.51 mA.

(c) I_{B2} flows *out* of the pnp base into node C1. KCL at C1: (current down through R_{C1}) + I_{B2} = I_{C1}, so R_{C1} carries $I_{C1} - I_{B2}$. With $I_{B2} = I_{E2}/101$ and $I_{E2} = (12 - V_{C1} - 0.7)/2.2$:

$$V_{C1} = 12 - 4.7\left(1.322 - \frac{11.3 - V_{C1}}{222.2}\right) \Rightarrow V_{C1} = \mathbf{5.90\ V}$$

Then V_{E2} = 6.60 V, I_{E2} = 2.454 mA, I_{B2} = 24.3 µA and $I_{C2} = 100 I_{B2}$ = 2.429 mA.

$$V_{C2} = 2.429 \times 1.8 = \mathbf{4.37\ V}$$

Checks: Q_2: V_{EC2} = 6.60 − 4.37 = 2.23 V > 0.2 V, so active. Q_1: V_{E1} = 1.33 V and V_{CE1} = 5.90 − 1.33 = 4.57 V > 0.2 V, so active. Neglecting I_{B2} changed V_{C1} by only 0.11 V, because I_{B2} is 2 % of I_{C1}.

(d) At the edge, $V_{C2} = V_{E2} - 0.2$ = 6.40 V with the same I_{C2}, so $R_{C2,\max}$ = 6.40/2.429 = **2.64 kΩ**. With a 3.3 kΩ resistor, Q_2 would saturate.

New Q5 [8 marks] (design).

(a) V_E = 1.5 V. $R_E = V_E/I_E \approx 1.5/2$ = **0.75 kΩ**. $R_C = (15 - 6 - 1.5)/2$ = **3.75 kΩ**.

(b) $R_2 \le \beta_{\min} R_E/10 = 100(0.75)/10$ = **7.5 kΩ**. $V_B = 1.5 + 0.7$ = 2.2 V, so $R_1 = R_2(V_{CC} - V_B)/V_B = 7.5(12.8)/2.2$ = **43.6 kΩ**.

(c) Choose R_1 = 43 kΩ, R_2 = 7.5 kΩ, R_E = 750 Ω and R_C = 3.6 kΩ. That gives R_{Th} = 6.39 kΩ and E_{Th} = 15(7.5/50.5) = 2.228 V.

β	$I_B = \dfrac{2.228 - 0.7}{6.39 + (\beta+1)0.75}$	I_{CQ}	$V_{CEQ} = 15 - I_C(3.6) - I_E(0.75)$
100	18.6 µA	**1.86 mA**	**6.90 V**
300	6.58 µA	**1.97 mA**	**6.41 V**

Tripling β changes I_C by only 6 %. The 3.9 kΩ standard value would also work: 6.34 V and 5.81 V. Both sit close to the 6 V target.

New Q6 [6 marks] (collector feedback, β range).

(a) R_C carries I_E. Loop from 12 V through R_C, R_F, V_{BE} and R_E:

$$I_B = \frac{12 - 0.7}{180 + (\beta+1)(3.3 + 0.5)}, \qquad V_C = 12 - 3.3 I_E, \qquad V_{CE} = 12 - 3.8 I_E$$

β	I_B	I_C	V_C	V_{CE}
80	23.2 µA	1.85 mA	5.81 V	4.87 V
160	14.3 µA	2.28 mA	4.42 V	3.27 V
240	10.3 µA	2.47 mA	3.80 V	2.56 V

All three are active. Tripling β raises I_C by 34 %.

(b) $I_B = 2.283/160$ = 14.27 µA, so $R_B = 11.3/0.01427$ = **792 kΩ**. At β = 240, $I_C = 240(11.3/792)$ = 3.43 mA and $V_{CE} = 12 - 3.43(3.3)$ = **0.70 V**, which is almost saturated. At β = 80 it is 1.14 mA and 8.23 V. With fixed bias, I_C rises in proportion to β (×3). With collector feedback, a rising I_C lowers V_C, which reduces I_B. See the figure in Section 1.3.

6. Quick recall check

Question

1. Saturation. Use $V_{CE} = 0.2$ V (with $V_{BE} = 0.7$ V) and solve the base and collector loops together with $I_E = I_B + I_C$. Then check that $I_C < \beta I_B$.
2. $\beta_{\max}$ = 200. It gives the largest I_C and so the smallest V_{CE}, which is the case closest to saturation.
3. $I_E = I_C + I_B$, because R_F's current (= I_B) also leaves the collector node through R_C.
4. V_B = 2.3 V. I_B and I_C both flow *out* of the transistor; I_E flows in.
5. $V_E \approx 0.1 V_{CC}$ (so $R_E = V_E/I_C$), and $R_2 \leq \beta_{\min} R_E/10$.